

HEMLING, Sweden

WMO: 022650

Lat: 63.6503N

Lon: 18.5466E

Elev: 183

StdP: 99.15

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-22.6	-28.7	0.3	-25.9	-25.0	0.4	-22.5	7.9	-0.2	7.1	-0.6	0.5	340	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	25.3	17.2	23.3	16.3	21.6	15.5	18.6	23.0	17.5	21.8	16.3	20.1	3.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.5	20.0	15.7	11.4	18.8	14.7	10.6	17.7	53.4	22.8	49.8	22.0	46.3	20.1	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.1	5.3	DB	-29.0	27.8	3.1	2.1	-31.2	29.3	-33.0	30.6	-34.7	31.7	-37.0	33.3	(4)
(5)			WB	-29.3	20.1	3.0	1.3	-31.4	21.0	-33.2	21.8	-34.9	22.6	-37.0	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-8.7	-8.0	-4.0	1.6	7.5	11.7	14.9	13.0	8.4	2.2	-2.5	-6.3	(6)	
	DBStd	9.30	6.58	6.69	5.11	3.42	3.65	3.06	2.76	2.92	3.08	3.81	4.75	6.32	(7)	
	HDD10.0	3076	581	503	433	251	96	17	1	7	66	242	375	506	(8)	
	HDD18.3	5771	839	737	691	501	336	201	113	167	298	499	625	764	(9)	
	CDD10.0	354	0	0	0	0	18	67	151	101	17	1	0	0	(10)	
	CDD18.3	7	0	0	0	0	0	1	5	1	0	0	0	0	(11)	
	CDH23.3	172	0	0	0	0	7	35	103	27	0	0	0	0	(12)	
	CDH26.7	20	0	0	0	0	0	4	15	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.1	1.9	1.9	2.4	2.3	2.5	2.6	2.2	2.0	2.1	2.0	1.9	1.9	(14)	
(15) Precipitation	PrecAvg	653	48	31	30	32	41	52	90	95	63	60	57	52	(15)	
	PrecMax	940	88	73	86	74	70	113	215	177	160	162	142	125	(16)	
	PrecMin	422	5	4	8	4	4	21	2	18	12	17	17	4	(17)	
	PrecStd	123	20	16	16	20	18	21	47	48	41	32	36	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.4	6.3	9.2	18.1	24.4	27.1	28.7	26.1	19.8	12.9	8.7	5.3	(19)	
		MCWB	3.5	3.4	5.0	9.5	14.5	16.4	18.9	18.6	15.0	9.8	7.9	3.8	(20)	
	2%	DB	3.1	5.0	7.0	13.8	21.2	23.7	26.0	23.7	17.6	11.0	6.4	4.0	(21)	
		MCWB	2.0	2.8	3.5	8.0	13.0	15.3	18.0	17.3	13.3	9.3	5.9	2.7	(22)	
	5%	DB	1.6	2.7	5.4	10.8	18.5	21.1	24.0	21.7	15.7	9.6	4.8	2.6	(23)	
		MCWB	0.7	1.1	2.4	5.8	11.6	13.9	17.1	16.2	12.2	8.4	4.2	1.8	(24)	
	10%	DB	0.2	0.8	3.6	8.5	15.9	19.2	22.2	19.6	14.3	8.4	3.6	1.5	(25)	
		MCWB	-0.2	0.0	1.3	4.3	10.1	13.1	16.4	15.4	11.8	7.3	3.1	0.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.9	5.4	10.5	15.6	17.7	20.3	19.9	15.6	10.8	8.0	4.2	(27)	
		MCDB	5.1	5.7	8.8	17.2	21.4	24.4	24.8	23.8	18.5	11.7	8.4	5.1	(28)	
	2%	WB	2.1	2.9	3.9	8.1	14.0	16.1	19.1	18.3	14.2	9.7	6.1	2.9	(29)	
		MCDB	3.0	4.8	6.5	13.6	20.6	21.9	23.5	21.9	16.3	10.6	6.4	3.5	(30)	
	5%	WB	0.8	1.4	2.7	6.1	11.9	14.9	18.0	17.0	13.2	8.6	4.3	1.9	(31)	
		MCDB	1.4	2.5	4.9	10.0	17.6	19.5	22.8	20.4	14.9	9.5	4.7	2.5	(32)	
	10%	WB	-0.1	0.1	1.5	4.6	10.4	13.7	17.0	15.8	12.2	7.4	3.1	1.0	(33)	
		MCDB	0.3	0.7	3.3	8.0	15.2	18.0	21.2	18.7	13.8	8.3	3.5	1.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	9.1	10.1	10.4	11.6	11.3	11.3	10.8	9.7	7.5	5.5	7.3	(35)	
		MCDBR	8.3	8.4	11.0	15.5	17.2	15.3	15.2	14.9	12.4	9.6	5.7	6.9	(36)	
		MCWBR	7.7	7.1	8.0	9.5	9.7	7.9	7.9	8.7	8.5	7.6	5.2	6.4	(37)	
	5% WB	MCDBR	7.8	7.9	10.2	14.5	16.1	13.9	13.1	12.7	10.4	8.7	5.5	6.6	(38)	
		MCWBR	7.4	6.9	7.6	9.2	9.4	7.7	7.5	7.9	7.7	7.3	5.2	6.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.214	0.278	0.305	0.343	0.344	0.342	0.358	0.347	0.316	0.301	0.360	0.350	(40)	
	taud		2.022	2.191	2.266	2.338	2.450	2.483	2.457	2.479	2.555	2.434	2.726	2.673	(41)	
	Ebn at Noon		389	646	788	836	870	878	851	828	791	635	368	213	(42)	
	Edh at Noon		28	60	84	98	96	95	95	85	65	48	25	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.15	0.75	2.24	3.78	4.89	5.53	5.09	3.86	2.35	0.95	0.24	0.07	(44)	
	RadStd		0.02	0.14	0.24	0.34	0.56	0.38	0.52	0.34	0.27	0.17	0.05	0.01	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-127	N/A	N/A	N/A	

Nomenclature: See separate page